

Foundation (Jalapeno)

- Q1.** What is the vertex of the graph of $y = |x|$?
 (A) (0, 0)
 (B) (0, 1)
 (C) (1, 0)
 (D) (-1, 0)
 (E) (1, 1)
- Q2.** Which of the following is the graph of $y = |x - 2|$?
 (A) A graph with a vertex at (0, 2)
 (B) A graph with a vertex at (2, 0)
 (C) A graph with a vertex at (2, 2)
 (D) A graph with a vertex at (-2, 0)
 (E) A graph with a vertex at (0, 0)
- Q3.** What is the axis of symmetry of the graph of $y = |x + 1|$?
 (A) $x = -1$
 (B) $x = 0$
 (C) $x = 1$
 (D) $y = -1$
 (E) $y = 0$
- Q4.** What is the direction of opening of the graph of $y = |x|$?
 (A) Up
 (B) Down
 (C) Left
 (D) Right
 (E) None of the above
- Q5.** What is the graph of $y = 2|x|$?
 (A) A graph that is narrower than the graph of $y = |x|$
 (B) A graph that is wider than the graph of $y = |x|$
 (C) A graph that is the same as the graph of $y = |x|$
 (D) A graph with a vertex at (0, 2)
 (E) A graph with a vertex at (2, 0)
- Q6.** What is the graph of $y = |x| - 1$?
 (A) A graph that is the same as the graph of $y = |x|$ but shifted down 1 unit
 (B) A graph that is the same as the graph of $y = |x|$ but shifted up 1 unit
 (C) A graph that is the same as the graph of $y = |x|$ but shifted left 1 unit
 (D) A graph that is the same as the graph of $y = |x|$ but shifted right 1 unit
 (E) A graph with a vertex at (0, -1)
- Q7.** What is the equation of the graph that is a reflection of the graph of $y = |x|$ across the x-axis?
 (A) $y = -|x|$
 (B) $y = |x|$
 (C) $y = -x$
 (D) $y = x$
 (E) $y = |x| + 1$
- Q8.** Which of the following is a transformation of the graph of $y = |x|$?
 (A) A translation of 2 units to the right
 (B) A translation of 2 units to the left
 (C) A reflection across the y-axis
 (D) A vertical stretch by a factor of 2
 (E) All of the above

Open Ended Questions (FRQ)

- Q9.** Graph the function $y = |x - 2| + 1$ and identify its vertex, axis of symmetry, and direction of opening.

- Q10.** Compare and contrast the graphs of $y = |x|$ and $y = |x| - 2$. How do the transformations affect the vertex, axis of symmetry, and direction of opening?

Chef's Tips

- Tip 1: Use graph paper to help students visualize the transformations.
- Tip 2: Encourage students to use the properties of absolute value functions to identify the vertex, axis of symmetry, and direction of opening.
- Tip 3: Remind students to check their work by plugging in values for x to ensure their graph is correct.